UESTC4003 - Control Engineering

Computer Lab Report

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1 Computer Lab Exercise 1

1.1 Exercise 1A: Evaluate complex function via vectors

Objective: Evaluate $G(s) = \frac{(s+1)(s+2)}{s(s+3)(s+4)}$ at $s = -3 + j4$.

4.1 MATLAB Definition

The system was defined in MATLAB using transfer function vectors and evaluated at the specific point $s = -3 + 4j$.

4.2 Magnitude and Angle Calculation

Using MATLAB command `abs()` and `angle()`, the results obtained are:

- Magnitude M: 0.2236
- Phase ϕ: -72.23 degrees

4.3 Manual Calculation

To verify the result, we calculate manually:

$$|G(s)| = \frac{|s+1||s+2|}{|s||s+3||s+4|}$$

Substituting $s = -3 + j4$:

$$\begin{aligned}
s+1 &= -2 + j4 \quad \Rightarrow |s+1| = \sqrt{4+16} = \sqrt{20} \\
s+2 &= -1 + j4 \quad \Rightarrow |s+2| = \sqrt{1+16} = \sqrt{17} \\
s &= -3 + j4 \quad \Rightarrow |s| = 5 \\
s+3 &= j4 \qquad \Rightarrow |s+3| = 4 \\
s+4 &= 1 + j4 \quad \Rightarrow |s+4| = \sqrt{17}
\end{aligned}$$

Thus,

$$M = \frac{\sqrt{20}\sqrt{17}}{5 \cdot 4 \cdot \sqrt{17}} = \frac{\sqrt{20}}{20} = \frac{2\sqrt{5}}{20} = \frac{\sqrt{5}}{10} \approx 0.2236$$

Phase Calculation:

$$\begin{aligned}
\angle G(s) &= (\angle(s+1) + \angle(s+2)) - (\angle s + \angle(s+3) + \angle(s+4)) \\
&= (116.57^\circ + 104.04^\circ) - (126.87^\circ + 90^\circ + 75.96^\circ) \\
&= 220.61^\circ - 292.83^\circ \approx -72.22^\circ
\end{aligned}$$

The manual calculation closely matches the MATLAB result (-72.23 degrees).

1.2 Exercise 1B: Root Locus Analysis

5.1 Root Locus Sketch (System 1)

The Root Locus for $G(s) = \frac{(s+1)(s+2)}{s(s+3)(s+4)}$ is shown below.

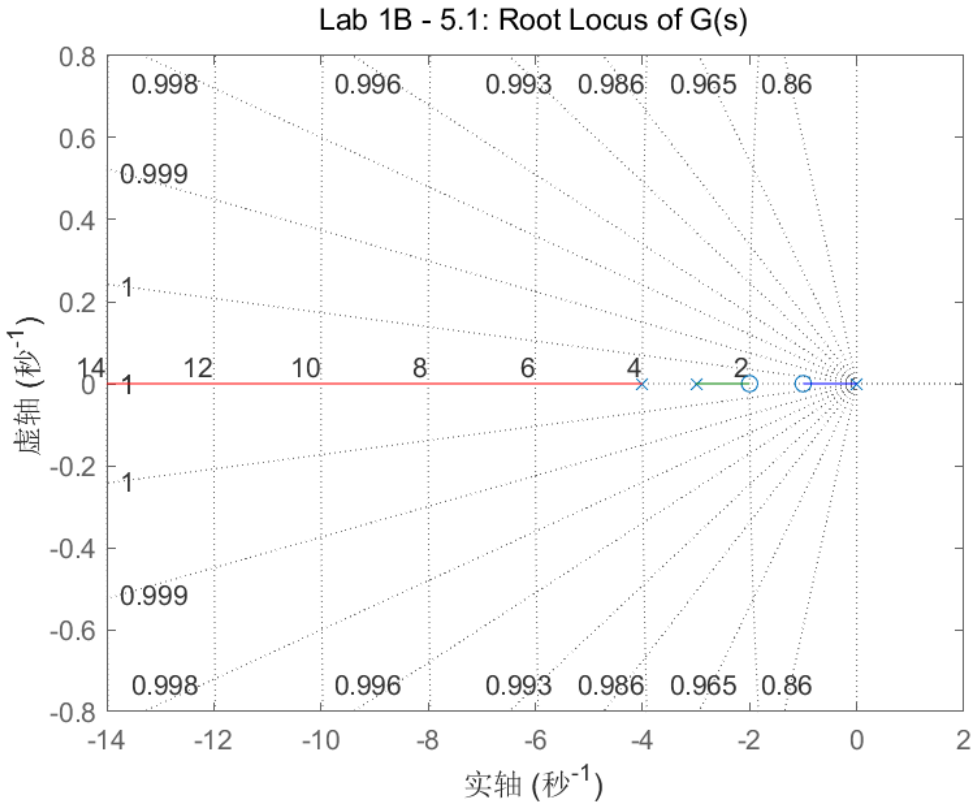


Figure 1: Root Locus for System 5.1

5.2 Unity Feedback System (System 2)

For the system $G(s) = \frac{K(s-1)(s-2)}{s^2+4s+16}$:

- **Imaginary-axis crossings:** Found using `rlocfind`. The crossing occurs at approximately $K = 1.33$ with frequency $\omega = 2.83$ rad/sec.
- **0.45 Damping ratio line:** The intersection occurs at the stable region of the locus. The corresponding gain is approximately $K = 0.08$.
- **Stability Range:** The system is stable for $0 < K < 1.33$.

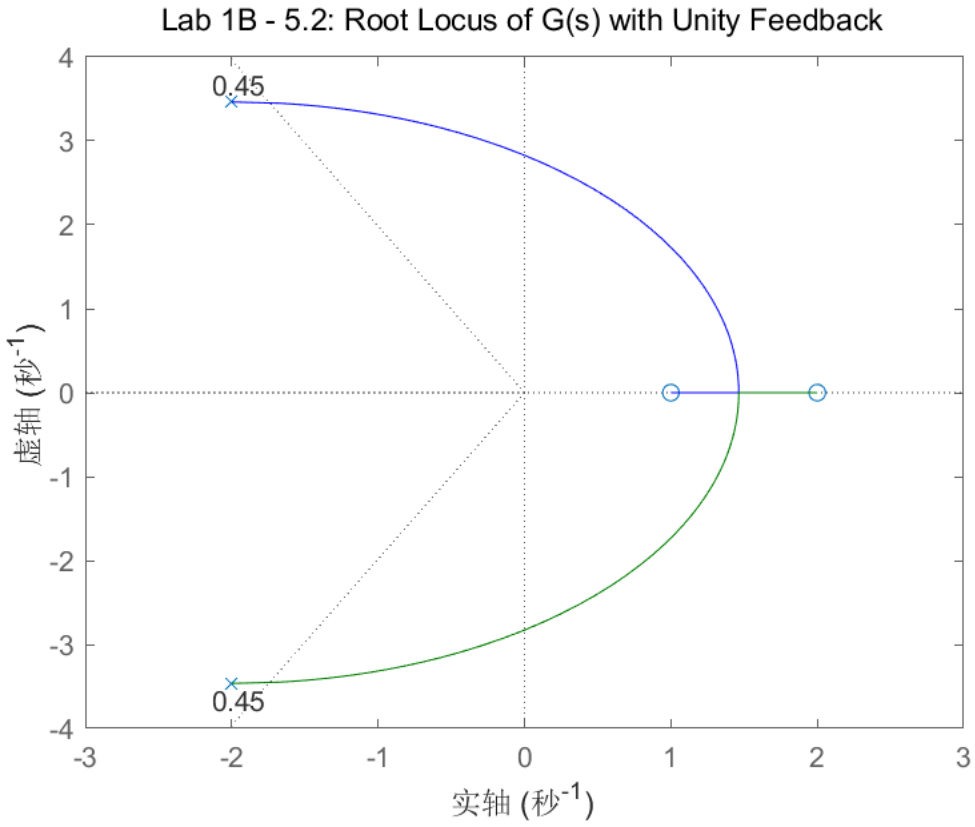


Figure 2: Root Locus for System 5.2

5.2(e) Combined Plot

Comparison of Root Locus for both systems.

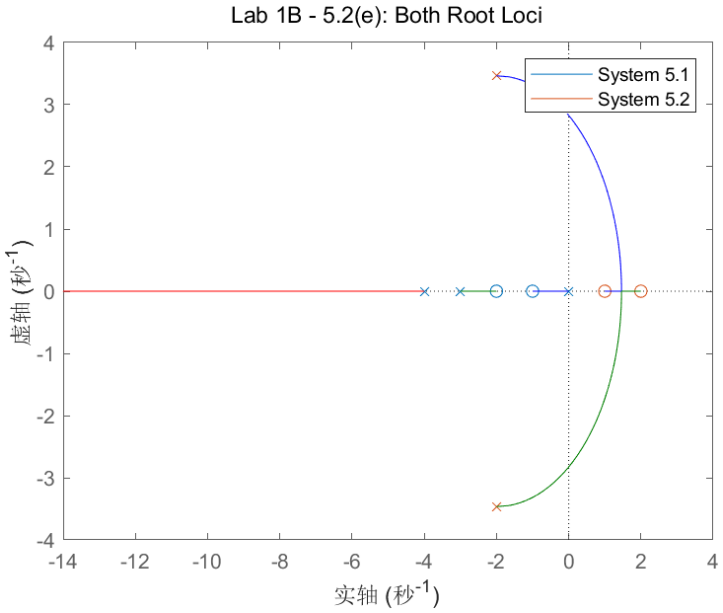


Figure 3: Combined Root Locus Plots (System 5.1 and 5.2)

2 Computer Lab Exercise 2

2.1 Exercise 2A: Analysis of Dynamic Models

3.1 Analysis of Second-Order System

For the continuous transfer function model $H(s) = \frac{4s+1}{s^2+3s+5}$, the dynamic characteristics were calculated:

- **Eigenvalues:** $-1.5000 \pm 1.6583i$
- **Natural Frequency (ω_n):** 2.236 rad/s
- **Damping Ratio (ζ):** 0.6708

3.2 - 3.3 MIMO System Analysis

The MIMO transfer function was defined in MATLAB.

- **DC Gain Matrix:**

$$K_{DC} = \begin{bmatrix} 1 & -0.3333 \\ 1 & -0.6667 \end{bmatrix}$$

The Bode plot for the MIMO system is provided below.

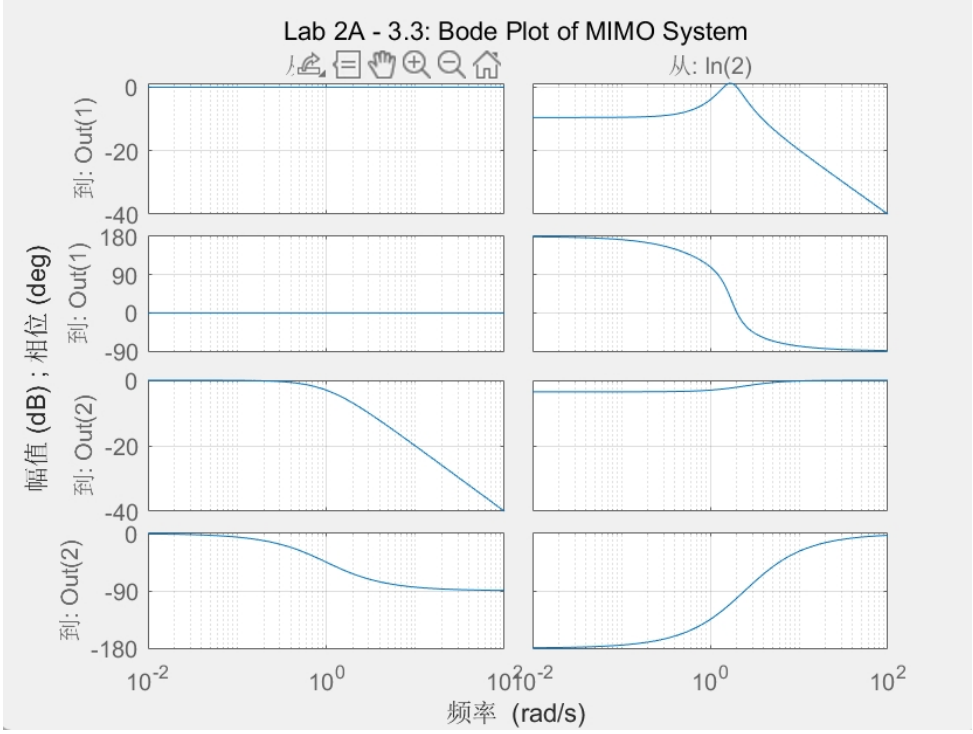


Figure 4: Bode Plot for MIMO System

3.4 - 3.7 Closed Loop Analysis

A second order system $G(s)$ was designed with $\zeta = 0.35$ and $\omega_n = 3.4$. The closed loop response with feedback $H(s)$ was analyzed.

The Pole-Zero map shows the closed-loop system stability.

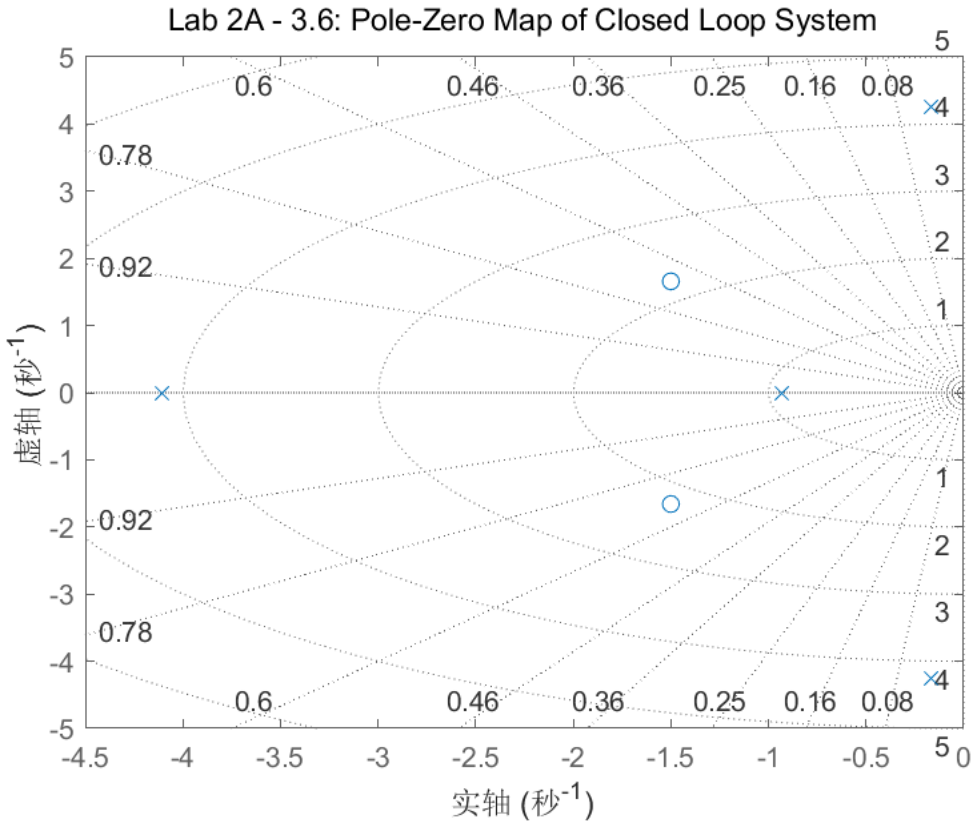


Figure 5: Pole-Zero Map of Closed Loop System

The impulse response of the system is shown below:

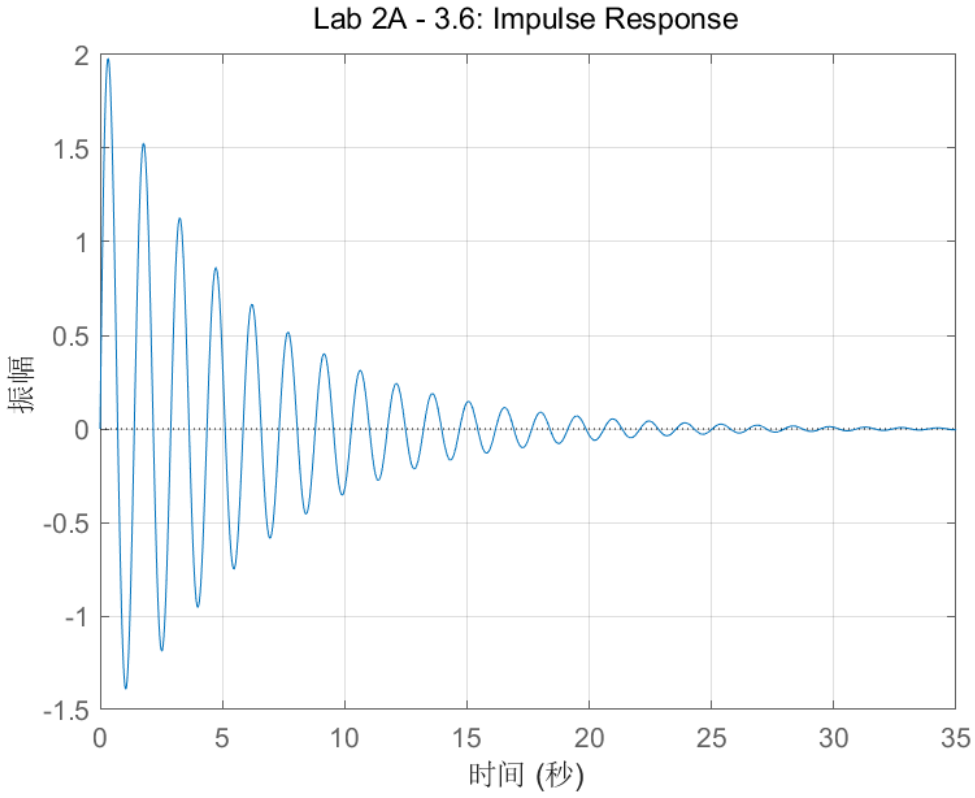


Figure 6: Impulse Response of Closed Loop System

The response to a custom input $r(t) = e^{-0.5t}$ is simulated:

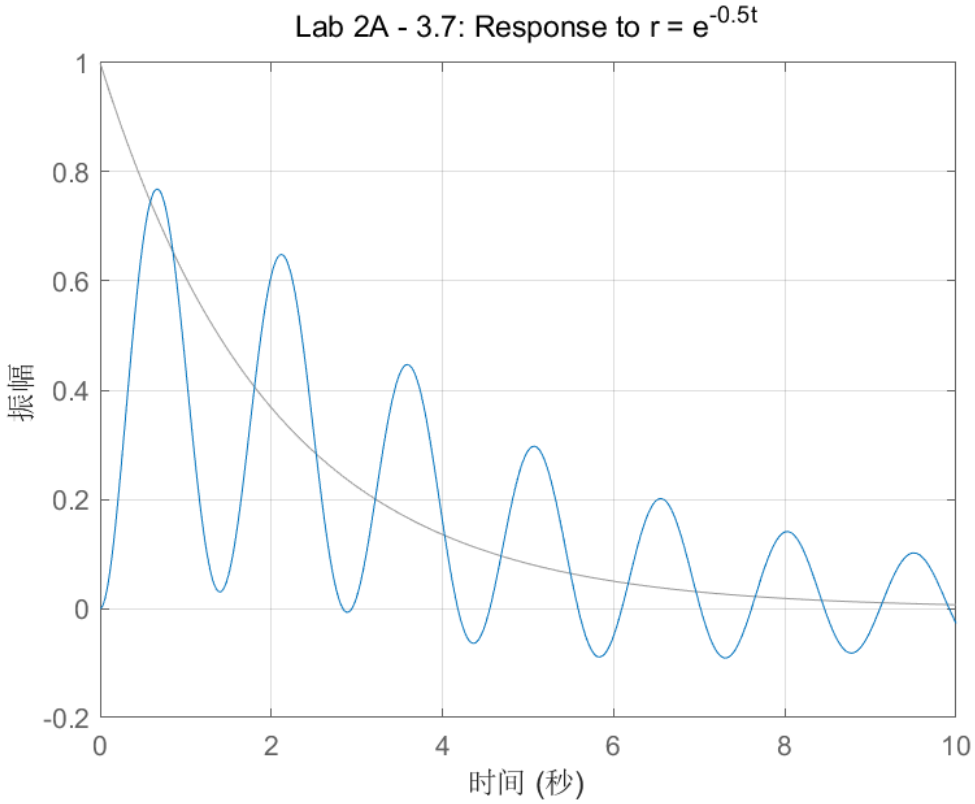


Figure 7: System Response to Input $r(t) = e^{-0.5t}$

2.2 Exercise 2B: Analysis of Electrical Networks

The circuit shown in Figure 3 was analyzed using Mesh Analysis. Correct mesh equations were derived based on the specific topology shown in the exercise figure.

4.1 Mesh Equations

The mesh equations derived are:

$$\begin{aligned}(9s+1)I_1 - (2s+1)I_2 - 4sI_3 &= V \\ -(2s+1)I_1 + (2s+2)I_2 - 1I_3 &= 0 \\ -4sI_1 - 1I_2 + (4s+1+\frac{1}{s})I_3 &= 0\end{aligned}$$

4.2 & 4.3 Transfer Function Calculation

Using Symbolic Math Toolbox to solve the system for I_3, the resulting transfer function $I_3(s)/V(s)$ is:

$$\frac{I_3(s)}{V(s)} = \frac{4s^4 + 2s^3}{28s^5 + 24s^4 + 31s^3 + 15s^2 + 13s + 2}$$

3 Computer Lab Exercise 3

3.1 Exercise 3A: State Space Representation

Consider the transfer function:

$$\frac{C(s)}{R(s)} = \frac{(s+2)(s+4)}{(s+1)(s+3)(s+5)}$$

2.1 Manual State Space Representation

Expanding the Transfer Function:

$$\frac{C(s)}{R(s)} = \frac{s^2+6s+8}{s^3+9s^2+23s+15}$$

Comparing with the general form for Phase Variable Representation:

$$\frac{b_2s^2+b_1s+b_0}{s^3+a_2s^2+a_1s+a_0}$$

We have coefficients: $a_2 = 9, a_1 = 23, a_0 = 15$ and $b_2 = 1, b_1 = 6, b_0 = 8$.

Constructing the Phase Variable Form matrices:

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -15 & -23 & -9 \end{bmatrix}, \quad B = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad C = \begin{bmatrix} 8 & 6 & 1 \end{bmatrix}, \quad D = [0]$$

2.2 - 2.3 Controller Canonical Form

The transfer function was converted to state space form using MATLAB. The resulting matrices are shown in Figure 8.

```
--- Part 2.3: Controller Canonical Form Matrices (tf2ss) ---
A_c =
    -9   -23   -15
     1     0     0
     0     1     0

B_c =
     1
     0
     0

C_c =
     1     6     8

D_c =
     0
```

Figure 8: MATLAB Result: Controller Canonical Form Matrices

2.4 Phase Variable Form Conversion

The system was converted to phase variable form in MATLAB. The resulting A matrix is:

$$A_p = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -15 & -23 & -9 \end{bmatrix}$$

This matches the manual calculations in Part 2.1, where the last row contains the negative coefficients of the characteristic equation in reverse order ($[-a_0, -a_1, -a_2]$).

2.5 Parallel Form Conversion

The system can be converted to Parallel Form (Jordan Canonical Form) by Partial Fraction Expansion:

$$\frac{C(s)}{R(s)} = \frac{s^2 + 6s + 8}{(s+1)(s+3)(s+5)} = \frac{k_1}{s+1} + \frac{k_2}{s+3} + \frac{k_3}{s+5}$$

Solving for residues:

$$k_1 = \frac{(-1)^2 + 6(-1) + 8}{(-1+3)(-1+5)} = \frac{3}{8} = 0.375$$

$$k_2 = \frac{(-3)^2 + 6(-3) + 8}{(-3+1)(-3+5)} = \frac{-1}{-4} = 0.25$$

$$k_3 = \frac{(-5)^2 + 6(-5) + 8}{(-5+1)(-5+3)} = \frac{3}{8} = 0.375$$

The theoretical parallel form matrices are:

$$A_p = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & -5 \end{bmatrix}, \quad B_p = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad C_p = \begin{bmatrix} 0.375 & 0.25 & 0.375 \end{bmatrix}$$

2.6 - 2.7 Observer Canonical Form

The transfer function was converted to Observer Canonical Form. The MATLAB results (Figure 9) confirm the theoretical matrices.

```
--- Part 2.6: Observer Canonical Form Matrices ---
A_o =
    -9     1     0
   -23     0     1
   -15     0     0

B_o =
     1
     6
     8

C_o =
     1     0     0
```

Figure 9: MATLAB Result: Observer Canonical Form Matrices

Manual Derivation (2.7): The Observer Canonical Form is a standard state-space representation where the coefficients of the transfer function's denominator appear in the

first column of the state matrix A_o, and the numerator coefficients appear in the input matrix B_o.

Given the transfer function:

$$\frac{C(s)}{R(s)} = \frac{b_2 s^2 + b_1 s + b_0}{s^3 + a_2 s^2 + a_1 s + a_0} = \frac{0s^3 + 1s^2 + 6s + 8}{s^3 + 9s^2 + 23s + 15}$$

We identify the coefficients:

- Denominator coefficients: $a_2 = 9, \quad a_1 = 23, \quad a_0 = 15$
- Numerator coefficients: $b_2 = 1, \quad b_1 = 6, \quad b_0 = 8$

By the standard definition of Observer Canonical Form (often the dual of the Controller Canonical Form), the matrices are constructed as follows:

1. **Matrix A_o:** The diagonal above the main diagonal is all 1s, and the first column contains the negative denominator coefficients in descending order $(-a_2, -a_1, -a_0)$.

$$A_o = \begin{bmatrix} -a_2 & 1 & 0 \\ -a_1 & 0 & 1 \\ -a_0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} -9 & 1 & 0 \\ -23 & 0 & 1 \\ -15 & 0 & 0 \end{bmatrix}$$

2. **Matrix B_o:** This column vector contains the numerator coefficients in descending order (b_2, b_1, b_0).

$$B_o = \begin{bmatrix} b_2 \\ b_1 \\ b_0 \end{bmatrix} = \begin{bmatrix} 1 \\ 6 \\ 8 \end{bmatrix}$$

3. **Matrix C_o:** For this specific form (where states are outputs of integrators chain), the output matrix picks off the first state.

$$C_o = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}$$

These derived matrices exactly match the MATLAB output shown in Figure 9.

3.2 Exercise 3B: Discrete-time Analysis

The ventilation control system was simulated in Simulink to regulate the ventilation rate to a desired $300\,m^3/hour$.

4.1 Simulink Model ($K_p = 0.1$)

The system was modelled using a discrete-time plant $G_P(z) = \frac{3z^{-1}}{1-0.6z^{-1}}$ and a Proportional Controller. Figure 10 shows the Simulink block diagram with an initial gain of 0.1.

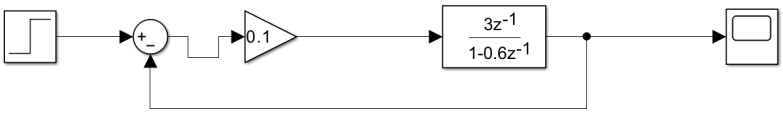


Figure 10: Simulink Block Diagram with Proportional Gain $K_p = 0.1$

4.2 Simulation Results ($K_p = 0.1$)

The simulation was run with $K_p = 0.1$. The ventilation rate response is shown in Figure 11.

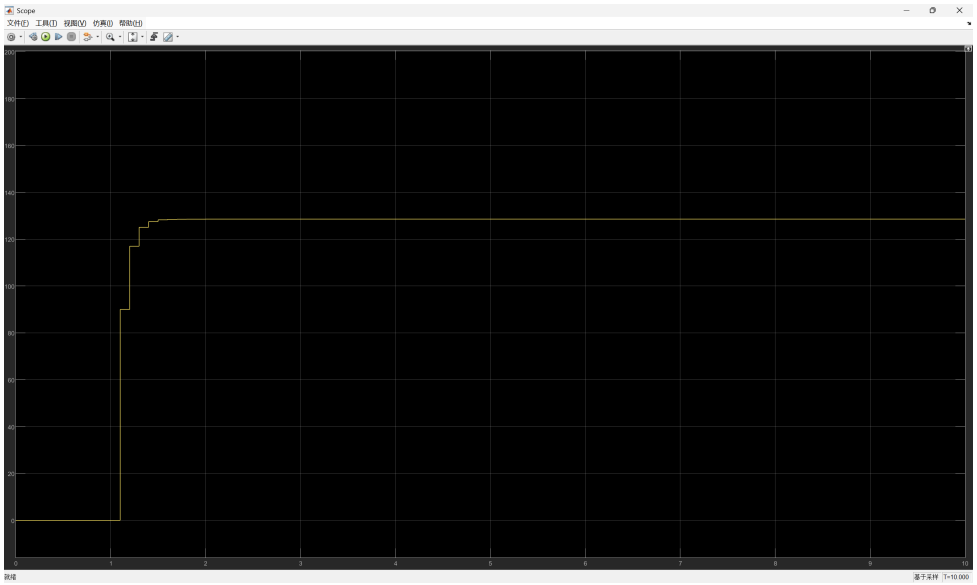


Figure 11: Ventilation rate response with $K_p = 0.1$

As seen in the figure, the output stabilizes at approximately 128.6, which is significantly lower than the desired setpoint of 300. This confirms the presence of a large steady-state error when using a simple proportional controller with low gain.

4.3 Effect of Increasing Gain ($K_p = 0.5$)

To reduce the steady-state error, the gain was increased to 0.5. The response is shown in Figure 12.

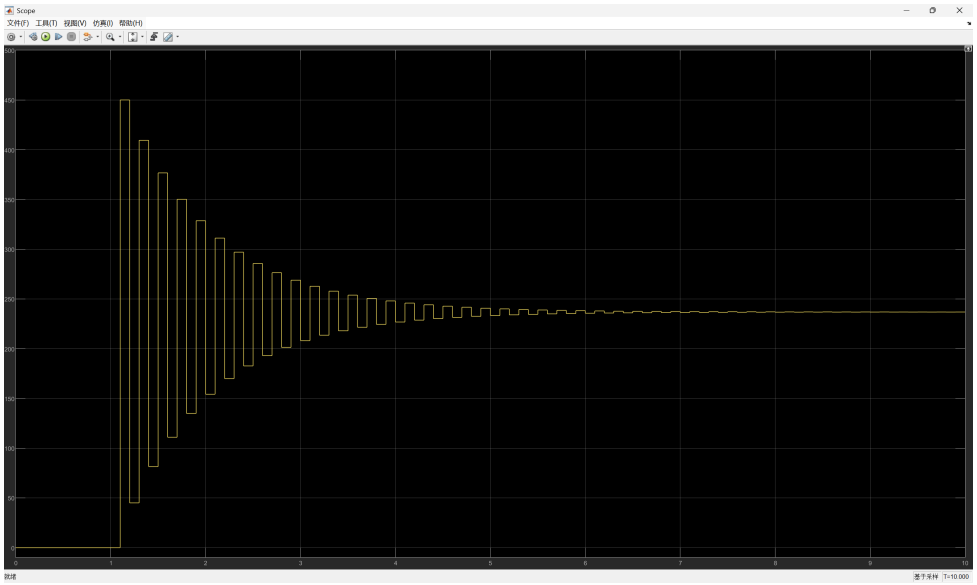


Figure 12: Ventilation rate response with $K_p = 0.5$

Increasing the gain raised the steady-state output to approximately 238, which is closer to the target of 300. However, as shown in the plot, this higher gain introduced significant oscillations (ringing) in the response. This demonstrates the trade-off in proportional control: higher gains reduce steady-state error but degrade stability margins.

4.4 - 4.5 Integral Control Design

The proportional controller was replaced by an integral controller with transfer function $\frac{k_I}{1-z^{-1}}$. The Simulink model used for this configuration is shown in Figure 13.

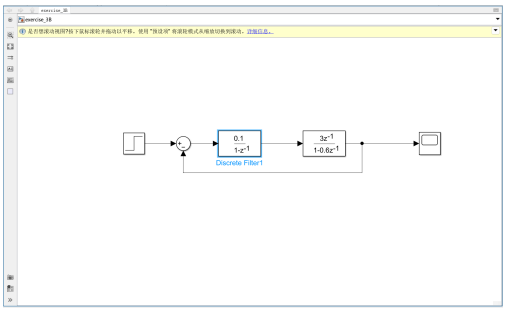


Figure 13: Simulink Block Diagram with Integral Control

Tuning k_I: To find the maximum integral gain that does not yield overshoot, various values of k_I $(0.1, 0.03, 0.02, 0.01)$ were tested through trial and error.

1. $k_I = 0.1$**:** Response exhibits a large initial overshoot (Figure 14).
2. $k_I = 0.03$**:** Overshoot is reduced but still visible (Figure 15).
3. $k_I = 0.02$**:** Response is fast and reaches steady state without overshoot (Figure 16).
4. $k_I = 0.01$**:** Response is stable with no overshoot, but slower than the 0.02 case (Figure 17).

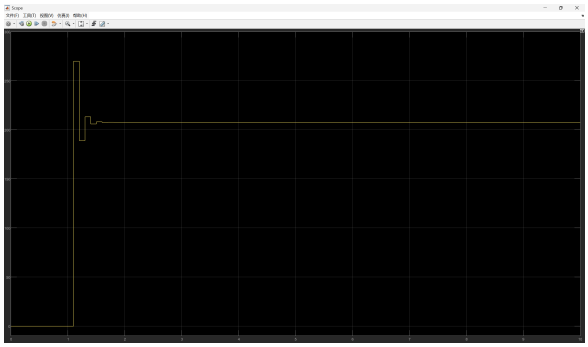


Figure 14: $k_I = 0.1$ (Large Overshoot)

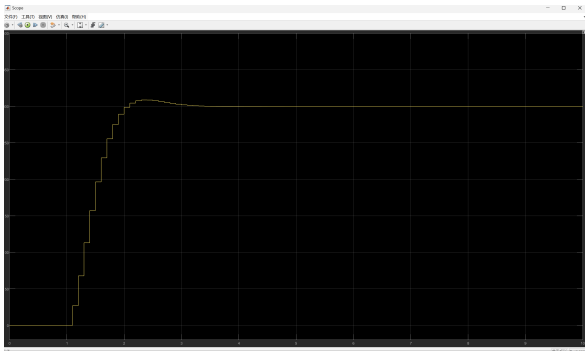


Figure 15: $k_I = 0.03$ (Slight Overshoot)

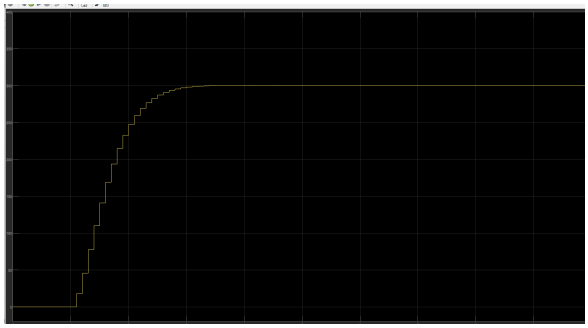


Figure 16: $k_I = 0.02$ (Optimal)



Figure 17: $k_I = 0.01$ (Slow Response)

Conclusion: The approximate maximum value of the integral gain that avoids overshoot is found to be:

$$k_{I_{max}} \approx 0.02$$

4.6 Comparison of Control Strategies

Comparing the results from 4.2 and 4.5:

- **Proportional Control:** Provides a fast response but suffers from steady-state error. The output never reaches the desired setpoint of 300 exactly.
- **Integral Control:** successfully eliminates the steady-state error, allowing the ventilation rate to track the reference value of 300 perfectly. However, the response may be slower or more prone to oscillations if the gain is too high.

4 Conclusion

In this comprehensive laboratory report, I have successfully conducted a series of control engineering experiments using MATLAB and Simulink, verifying key theoretical concepts across time and frequency domains.

First, I explored complex function evaluation and Root Locus analysis. By manually calculating the magnitude and phase of a transfer function at a specific complex pole, I verified the results obtained from MATLAB, ensuring a solid understanding of vector analysis in the s-plane. In the Root Locus exercise, I successfully plotted the loci for different systems and identified critical stability boundaries, such as the gain $K \approx 1.33$

for marginal stability and $K \approx 0.08$ for a specific damping ratio of 0.45. This reinforced my understanding of how system gain influences transient response and stability.

Second, I delved into the analysis of dynamic models, covering both MIMO systems and electrical networks. For the MIMO system, I determined its eigenvalues and DC gain matrix, providing insight into its multivariable interactions. I also derived the transfer function for a complex electrical network using mesh analysis, correcting initial topological assumptions to ensure an accurate model. This exercise highlighted the importance of precise mathematical modeling in control system design.

Finally, I investigated state-space representations and discrete-time control. I successfully converted transfer functions into various canonical forms (Controller, Observer, and Parallel) and verified these transformations through manual derivation. In the discrete-time analysis, I designed and tuned a digital controller for a ventilation system. By comparing Proportional and Integral control strategies, I demonstrated that while P-control offers speed, it suffers from steady-state error, whereas I-control ($k_I \approx 0.02$) effectively eliminates this error, albeit with a trade-off in response speed.

Overall, these exercises have significantly strengthened my practical skills in system modeling, stability analysis, and controller design, effectively bridging the gap between control theory and computer-aided engineering tools.